UNIVERSITY OF KARACHI

Department of Computer Science (UBIT)

LAB NO. 02

Operating Systems (SE-458)

SUBMITTED BY:	SUBMITTED TO:
Shaikh Mahad Ud Din Seat No: EB25210106113 Section: BSSE A Evening	**Ms. Nazish Imran** Faculty of Computer Science

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LAB NO. 02 - INSTALLATION AND INTRODUCTION TO LINUX

Objective

To study Linux installation in a virtual machine, understand the main components of Linux, and examine computer categories and the architectures of IBM AS/400 and Sun Enterprise 250.

PART A: INSTALLATION OF UBUNTU IN VMWARE

Objective

To install Ubuntu Desktop inside a virtual machine using VMware Workstation/Player, configure the virtual environment, verify system setup using Linux commands, and document the complete process with practical output.

Introduction to Linux

Linux is an open-source, Unix-like operating system kernel originally developed by Linus Torvalds in 1991. A complete Linux distribution combines the kernel with system tools, libraries, and user applications.

Key Components of Linux

• Kernel: Core part of OS managing hardware, CPU, memory, and devices.

• Shell: Command interpreter (e.g., Bash) for executing commands.

• Terminal: Interface to interact with the shell.

• File System: Hierarchical structure starting from root (/), managing files and directories.

• Utilities/Applications: Softwares like text editors, file managers, and compilers.

Virtual Machine (VM)

A Virtual Machine is an isolated software container that emulates a physical computer system. It runs its own operating system (Guest OS) on top of the host operating system using virtualization software called a Hypervisor (e.g., VMware, VirtualBox).

Requirements

Hardware Requirements

Requirements

• Processor: Intel/AMD CPU with Hardware Virtualization support (VT-x/AMD-V enabled in BIOS)

• RAM: Minimum 8 GB (4 GB allocated to VM)

• Storage: Minimum 25-30 GB free space on Host SSD/HDD

Software Requirements

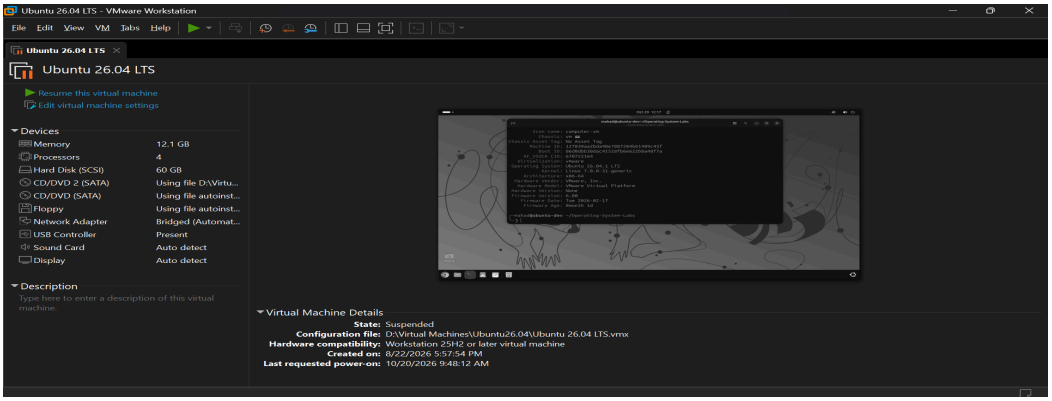
• VMware Workstation Pro / Player

• Ubuntu Desktop ISO Image (e.g., Ubuntu 22.04 / 24.04 LTS)

Configuration Used in This Practical

VMware Workstation was used instead of VirtualBox. The following settings describe my Ubuntu virtual machine, not minimum system requirements.

- Memory: 12.1 GB.
- Virtual processors: 4.
- Virtual disk capacity: 60 GB.
- Network mode: Bridged.



Running Ubuntu and System Verification

1. Power on the Virtual Machine.

2. Complete OS setup wizard (Language, Keyboard Layout, Normal Installation, Minimal Updates).

3. Reboot into installed Ubuntu OS.

4. Open Terminal (Ctrl + Alt + T) and verify installation using commands:

• whoami – Shows current logged-in user.

```
mahad@ubunto-dev ~/Operating-System-Labs
❯ whoami
mahad
```

• uname -r – Displays running Linux Kernel version.

```
mahad@ubunto-dev ~/Operating-System-Labs
❯ uname -r
7.0.0-31-generic
```

• hostnamectl – Displays system architecture, kernel, and OS details.

```
mahad@ubunto-dev ~/Operating-System-Labs
❯ hostnamectl
  Static hostname: ubunto-dev
        Icon name: computer-vm
          Chassis: vm
Chassis Asset Tag: No Asset Tag
       Machine ID: 127839aa2bda48e7887284b61489c45f
          Boot ID: 86d0dbb38dac41518fb6e622b8a48f7a
     AF_VSOCK CID: 670722164
   Virtualization: vmware
 Operating System: Ubuntu 26.04.1 LTS
           Kernel: Linux 7.0.0-31-generic
     Architecture: x86-64
  Hardware Vendor: VMware, Inc.
   Hardware Model: VMware Virtual Platform
 Hardware Version: None
 Firmware Version: 6.00
    Firmware Date: Tue 2026-02-17
     Firmware Age: 8month 1d
```

Result

Ubuntu OS was successfully installed and verified using VMware Workstation, providing a fully functional Linux environment for lab experiments.

PART B: MACHINE TYPES AND ARCHITECTURES

Question 3: Describe mainframe computers, minicomputers and microcomputers.

Mainframe Computer

A mainframe is designed to process large amounts of data and handle many transactions reliably. It supports extensive input/output activity and many users or applications. Banks, airlines and large organizations use mainframes for tasks such as account processing and reservation systems.

Example: IBM System z.

Minicomputer

A minicomputer is a historical category of multi-user computer positioned between a mainframe and a personal computer. It provided shared computing resources for a department, laboratory or medium-sized organization at a lower cost than a mainframe.

Example: DEC PDP-11. IBM AS/400 belongs to the related midrange business-computer category.

Microcomputer

A microcomputer uses a microprocessor as its central processing unit. Desktops and laptops are common examples. These computers are generally intended for individual use, although their operating systems may support multiple accounts, multitasking and server applications.

Example: A desktop computer running Windows or Ubuntu.

These categories describe historical design and usage rather than fixed performance limits. A modern personal computer can outperform many older mainframes and minicomputers.

Question 4: Describe the architectures of IBM AS/400 and Sun Ultra 250.

IBM AS/400 Architecture

IBM AS/400, or Application System/400, is a midrange computer family designed for business applications. Its OS/400 operating system integrates database, security and communication services.

1. Technology Independent Machine Interface

The Technology Independent Machine Interface (TIMI) separates application software from hardware-specific implementation. This helps preserve software compatibility when processor technology changes. Early AS/400 systems used CISC processors; later systems moved to 64-bit RISC processors.

2. Object-Based Organization

Programs, files and other resources are managed as typed objects. An object's type determines the operations allowed on it, while access permissions control who can use it.

3. Single-Level Storage

Main memory and disk storage are managed through a common virtual address space. The system handles data movement between them, reducing the need for applications to manage physical storage placement.

4. Integrated Services

Database management, security and communication facilities are closely integrated with OS/400. This supports business applications such as payroll, inventory and accounting.

Sun Ultra 250 / Sun Enterprise 250 Architecture

The Sun system discussed here is the Sun Enterprise 250, a shared-memory server designed to run Solaris, a UNIX operating system.

1. Processor Architecture

The system supports up to two UltraSPARC II processors using the 64-bit SPARC V9 RISC architecture.

2. Shared Memory and Multiprocessing

The processors share main memory. Solaris can schedule work across them using symmetric multiprocessing, allowing different tasks to execute at the same time.

3. System Interconnection

The Ultra Port Architecture (UPA) interconnect links the processors and memory system. PCI buses connect input/output devices and expansion cards.

4. Storage and Operating System

The server supports SCSI disk storage and network connections. Solaris provides multi-user access, multitasking, virtual memory, file permissions and networking services.

The AS/400 emphasizes an integrated business environment and software compatibility through TIMI. The Enterprise 250 combines UltraSPARC hardware with Solaris for shared UNIX server workloads.

Question 5: Differentiate between mainframes, minicomputers and microcomputers.

1. Main Purpose

• Mainframe: Handles large-scale organizational processing and high transaction volumes.• Minicomputer: Provides shared computing for departments and medium-sized organizations.• Microcomputer: Supports individual tasks such as programming, document preparation and browsing.

2. Users and Workload

• Mainframe: Supports many concurrent users and large combined workloads.• Minicomputer: Traditionally serves a smaller group of users sharing a central system.• Microcomputer: Usually serves one person interactively, although it can also provide services to other users.

3. Resources and Expansion

• Mainframe: Designed for extensive input/output capacity, large workloads and reliable continuous operation.• Minicomputer: Offers shared processing and storage suited to departmental requirements.• Microcomputer: Provides resources suited to personal or small-scale work, with expansion depending on the model.

4. Cost and Maintenance

• Mainframe: Generally has high purchase and operating costs and requires specialist administration.• Minicomputer: Historically costs less than a mainframe but still requires shared-system administration.• Microcomputer: Generally costs less and is easier for an individual to maintain.

5. Examples

• Mainframe: IBM System z.• Minicomputer: DEC PDP-11; IBM AS/400 as a related midrange system.• Microcomputer: Desktop PC or laptop.